Weekly Report 4: Determining Optimal Voxel Grid Size from Silhouette Dimensions and System Resources

Project Number: 18

1 Introduction

This week, we analyzed how to decide the **voxel grid size** for our 3D reconstruction system. A voxel grid defines the spatial sampling resolution of the 3D model. Choosing an appropriate grid size is crucial — a grid that is too coarse loses shape details, while one that is too dense becomes computationally expensive. We explored how voxel resolution can be determined based on both the **silhouette dataset dimensions** and available **system resources (CPU, RAM, GPU)**.

2 Methodology

Our decision process combined two practical considerations:

1. The **image dimensions and number of silhouettes**, which define the spatial information captured by the dataset.
2. The **computational capacity** of our hardware, which limits the number of voxels we can efficiently process.

2.1 Relation with Silhouette Dimensions

Each silhouette represents the projection of the object on a 2D plane. The resolution of these silhouettes determines how finely we can reconstruct the object's surface. In our dataset, each BMP image is of size approximately **512 × 512 pixels**, and there are **200 silhouettes** captured at 1.8° intervals.

To ensure the voxel grid corresponds well to this level of detail:

- The grid resolution should roughly match the scale of the silhouette — finer silhouettes justify a denser voxel grid.
- The number of silhouettes determines how much 3D coverage the object has; more silhouettes allow slightly coarser grids without major detail loss.
- Practically, a voxel grid between 64^3 and 128^3 provides sufficient sampling for images in the 512×512 range.

2.2 System Resource Consideration

Voxel grids are cubic in nature, meaning their size increases cubically with resolution (N^3). This directly affects both memory and processing time. Hence, it was essential to test how large a grid our current setup (8 GB RAM, standard CPU, no GPU) could handle without performance degradation.

We performed tests with different voxel grid resolutions and measured memory usage and processing times.

Implementation in Python:

```
import numpy as np, time

def create_voxel_grid(resolution=64):
    start = time.time()
    x = np.linspace(-1, 1, resolution)
    y = np.linspace(-1, 1, resolution)
    z = np.linspace(-1, 1, resolution)
    grid = np.stack(np.meshgrid(x, y, z, indexing='ij'), axis=-1)
    print(f"{resolution}³ -> Memory: {grid.nbytes/1024**2:.2f} MB, Time: {time.time()
    return grid
```

Observations:

- 32^3 grid: Fast but very low spatial resolution — shape boundaries become blocky.
- 64^3 grid: Balanced detail and performance — ideal for silhouettes up to 512×512 pixels.
- 128^3 grid: Higher accuracy but causes heavy memory usage and lag on CPU-only setups.

3 Results

By combining the above observations, we decided to use a voxel grid resolution of 64^3. This size preserves key structural details from the 512×512 silhouettes while maintaining smooth computation.

Table 1: Performance Analysis for Different Voxel Grid Sizes

Resolution	**Approx. Memory (MB)**	**Time (s)**	**Remarks**
32^3	2.1	0.2	Too coarse, poor shape detail
64^3	16.8	0.8	Balanced performance and accuracy
128^3	134.2	3.5	High detail, but exceeds CPU limits

4 Discussion

The voxel grid size must be carefully chosen to align with both the dataset and the computing hardware:

- The **silhouette image resolution** defines how finely we can model the object surface — higher-resolution images can justify denser voxel grids.
- The **number of silhouettes** determines how much angular coverage we have; fewer silhouettes may need denser grids to preserve object shape.
- The **system's CPU and memory** limit how large a voxel grid can be processed efficiently.

In our case, since we have 200 high-resolution silhouettes and moderate hardware, a 64^3 **grid** was the most practical choice. It captures sufficient detail for visual hull carving without exceeding memory limits.

5 Conclusion

This week, we concluded that voxel grid size should not be chosen arbitrarily. It must depend on:

- The **resolution and number** of silhouette images.
- The **computational power and memory** available for processing.

Through testing, we finalized a voxel resolution of 64^3 for further visual hull carving and mesh extraction. This ensures both accurate reconstruction and efficient execution in our upcoming stages.